

Question Paper

Exam Date & Time: 18-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIRST SEMESTER M.TECH. EXAMINATIONS - NOVEMBER 2025

Linear Algebra probability and statistics [MAT_5151]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Find the eigenvalues and eigenspace of a given matrix (4)

A) $A = \begin{bmatrix} 2 & -1 & 1 \\ 0 & 3 & 6 \\ 0 & 0 & 7 \end{bmatrix}$.

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Solve the following system of equations by the LU Decomposition method (4)

$$\begin{aligned} 3x_1 + x_2 + 6x_3 &= 0 \\ -6x_1 - 16x_3 &= 4 \\ 8x_2 - 17x_3 &= 17 \end{aligned}$$

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Let $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a transformation defined by (2)
 $T(x, y, z) = (x + 2y + 3z, 2y - 3x + z)$.
Determine whether T is a linear transformation or not.

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) Find the Singular Value Decomposition of the matrix (4)

A) $A = \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix}$

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) Find QR Decomposition of the matrix (3)

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 1 & 0 & 2 \\ 2 & -1 & 3 \end{bmatrix}$$

Course Outcome: "CO1" / Cognitive Level: "Application"

- C) Find the Moore-Penrose inverse of the matrix (3)

$$A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Course Outcome: "CO1" / Cognitive Level: "Application"

- 3) Let $\bar{X}$ be the mean of a sample of size 5 from a normal distribution with mean $\mu=0$, variance 125. Find k and c so that $P(\bar{X}<k)=0.67$ and $P(\bar{X}<c)=0.9960$, respectively. (5)

A)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) The lifetime of a certain brand of an electric bulb may be considered as a random variable with a mean of 1200 hours and a standard deviation of 250 hours. Find the probability using the Central Limit Theorem has the average lifetime of 60 bulbs exceeds 1250 hours. (5)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 4) The mean lifetime of a sample of 100 light tubes produced by a company is found to be 1580 hours. Test the hypothesis at level of significance that the mean lifetime of the tubes produced by the company 1600 hours with standard deviation of 90 hours. (5)

A)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) Find a linear regression equation for the following data: (5)

X	3	9	5	3	5	12
Y	8	6	4	2	9	7

Course Outcome: "CO3" / Cognitive Level: "Application"

- 5) Calculate the correlation coefficient for the following data: (5)

A)

X	100	200	300	400	500	600
Y	228	236	256	278	285	301

Course Outcome: "CO3" / Cognitive Level: "Application"

- B) Consider the mood of an individual modelled as a three-state Markov chain with the following transition probability matrix $P = \begin{bmatrix} 0.5 & 0.4 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{bmatrix}$. (5)

$$\text{probability matrix } P = \begin{bmatrix} 0.5 & 0.4 & 0.1 \\ 0.3 & 0.4 & 0.3 \\ 0.2 & 0.3 & 0.5 \end{bmatrix}$$

In the long run, what proportion of time does the process spend in each of the three states?

Course Outcome: "CO4" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-END SEMSTER-ODD-2025-PG MAT_5151:Linear Algebra probability and statistics

#	Course Outcome	Description	Marks
1	1	CO 1	20
2	2	CO 2	15
3	3	CO 3	10
4	4	CO 4	5
Total			50

* NA - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025-PG MAT_5151:Linear Algebra probability and statistics

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	7	26(52%)
4	Analysis	5	24(48%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	12	50 (100%)